Fiscal Year:	FY 2006	**Task Last Updated:**	FY 10/10/2006
PI Name:	Globus, Ruth Ph.D.		
Project Title:	Simulated Microgravity and Radiation-Induced Bone Degeneration: Oxidative Stress- and p53-Dependent Mechanisms		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation Biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	(1) **Bone Fracture**:Risk of Bone Fracture due to Spaceflight-induced Changes to Bone (2) **Cardiovascular**:Risk of Cardiovascular Adaptations Contributing to Adverse Mission Performance and Health Outcomes (3) **Osteo**:Risk Of Early Onset Osteoporosis Due To Spaceflight		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Organization Name:	NASA Ames Research Center		
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City:	Moffett Field	**State:**	CA
Zip Code:	94035-1000	**Congressional District:**	18
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2004 Radiation Biology NNH04ZUU005N
Start Date:	10/01/2005	**End Date:**	09/30/2009
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	3	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Changed Division and Discipline/Program to HRP as of FY2006, per program changes at that time, per JSC/A. Chu-ARC (jvp 4/2009)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Almeida, Eduardo A.C. (NASA Ames Research Center) Limoli, Charles (University of California Irvine) Searby, Nancy (NASA Ames Research Center) Vercoutere, Wenonah (NASA Ames Research Center)		
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Performance Goal Text:			

Task Description:	To determine the risks and mechanisms of tissue degeneration in the spaceflight environment, we will examine the combined effects of radiation and simulated weightlessness on bone. We will focus our studies on two important pathways, p53 and oxidative and nitrative stress-related species, in bone marrow progenitor populations because there is abundant evidence that the biological effects of weightlessness and those of radiation share these mechanism. Experiments will be conducted using the rodent hindlimb unloading model, the best validated animal model for simulating weightlessness, followed by irradiation. In Aim 1, we will test the hypothesis that simulated weightlessness of mice modulates the response of bone and bone marrow to radiation, both acutely and during tissue regeneration. In Aim 2 we will identify the target marrow-derived stem, progenitor and differentiated cell populations sensitive to the combined stimuli of simulated weightlessness and radiation. In Aim 3, we will explore the relevance of specific ROS/RNS and p53-dependent mechanisms for mediating cellular responses to combined simulated weightlessness and radiation. The results of these studies are expected to provide important information for the development of countermeasures both to help prevent degeneration of connective tissues such as bone during spaceflight and to improve subsequent tissue regeneration.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This project addresses several of the risks identified in NASA's Bioastronautics Critical Path Roadmap (BCPR) caused by long duration spaceflight. We will obtain basic new knowledge about bone tissue degeneration caused by musculoskeletal disuse and radiation at the molecular, cell, and tissue level, relevant to Earth, as well as in space. Results will also lead to the formulation of new countermeasures concepts and risk assessment strategies for bone and connective tissue degenerative diseases.
Task Progress:	The objectives of this research project are two-fold. First, to determine the combined effects of simulated weightlessness and radiation on connective tissue health and regeneration, and second, to define the relevant cellular and molecular mechanisms. The central hypothesis of this proposal is that the mechanical loading environment, as defined by gravity, influences cell responses to radiation via shared molecular pathways. Our specific aims are to: Aim 1. Test the hypothesis that simulated weightlessness modulates the response of bone and bone marrow to radiation, both acutely and during tissue regeneration. Aim 2 Determine the cellular basis for bone marrow-derived osteoblast and osteoclast responses to simulated weightlessness and radiation. Aim 3. Identify key molecular pathways mediating the cellular responses to simulated weightlessness and radiation. Hindlimb unloading is a standard experimental approach to simulate weightlessness on Earth. Hindlimb unloading causes osteopenia, reduced bone strength, reduced proliferation of osteoblast progenitors and increased numbers of osteoclasts ex vivo. We and others have found that hindlimb unloading of mice provides a test system that faithfully recapitulates many of the cardiovascular, hormonal, immunosuppressive, and musculoskeletal changes observed in astronauts and humans subjected to bedrest (the spaceflight analogue for humans. Once rodents are released from hindlimb unloading to ambulate normally, accretion of bone mass accelerates, tending to restore the structural integrity of bone, as is also observed in astronauts following return to Earth. Thus, release from hindlimb unloading provides the valuable opportunity to dissect mechanisms underlying tissue regeneration. Results from using this animal model, together with in vitro models developed to manipulate gravity-dependent loading such as hypergravity by centrifugation, suggest that proliferation and differentiation of cells are sensitive to gravity-dependent loading. Therefore, we implemented cell culture methods using bone marrow-derived cells to evaluate radiosensitivity of skeletal cell progenitors. We made significant progress during our initial year of funding using two experimental approaches, cell culture and animal studies, to address our major objectives. First, we developed a suite of cell culture methods and assays to test our hypotheses, acquiring baseline measurements that quantify the effects of irradiation alone and hypergravity alone on two different primary cell populations derived from the bone marrow of mice. Bone marrow stromal (BMS) cells matured under appropriate culture conditions from osteoprogenitors into osteoblast-like cells and were the primary focus of our first year's efforts. Also, haematopoietic cells were grown under specific conditions to mature into osteoclast-like cells, and effects of radiation and hypergravity on osteoclast-like cell and colony numbers were defined. Second, we began preliminary studies using animals subjected to hindlimb unloading or irradiation to start addressing our central hypothesis. Progenitor cells with potential to differentiate into either bone-forming osteoblasts or bone-resorbing osteoclasts were extracted from bone marrow of 6- 9 week old C57BL6/J male mice, then cells either were centrifuged at various doses and durations to simulate increased gravity loading (5-50 g for 5-180 min), or were exposed to a single dose of Ce-137 gamma irradiation (1-10 Gy at 1-10 Gy/min). Production of ROS was measured via fluorescence-activated flow cytometry by activation of the oxidation-sensitive dye, CM-H2DCFDA. Osteoblastogenesis was assessed by alkaline phosphatase activity and production of mineralized matrix (Alizarin red staining). We assessed BMS cell numbers following irradiation or centrifugation either by counting the cells (Coulter counter) or by measuring DNA content (CyQUANT assay). Osteoclastogenesis was assessed by manually counting those cells and colonies that stained positively for Tartrate Resistant Acid Phosphatase activity (TRAP+ cells). Our studies on BMS cells conclusively showed that both hypergravity and radiation increased oxidative stress in the form of elevated reactive oxygen species (ROS) when applied to cells cultured over 2-5 days after isolation. Interestingly, the increased ROS levels derived from either stimulus translated into divergent physiological effects. Whereas radiation treatments led to the expected cell kill, hypergravity resulted in a later increase in cell number. Baseline studies on haematopoietic cells (osteoclast precursors) showed that hypergravity increased the number of mono-nucleated and multi-nucleated cells expressing a biochemical marker of mature osteoclasts (Tartrate-Resistant Acid Phosphatase, TRAP). Conversely, irradiation caused the expected dose-dependent reduction in TRAP-positive colony and cell number. Thus, hypergravity increased numbers of both osteoblast-like and osteoclast-like cells, whereas irradiation exerted opposite effects; when observed in situ, these changes are indicative of an elevated rate of bone turnover, which we anticipate to be relevant to in vivo skeletal responses both to unloading and irradiation. Major scientific accomplishments are summarized below: 1) In vitro studies using BMS cells revealed both irradiation and hypergravity induce oxidative stress. 2) In vitro studies using BMS cells revealed irradiation (negative) and hypergravity (positive) have opposite effects on proliferation and differentiation. 3) These initial studies already generated data sufficient for a manuscript that was submitted to Radiation Biology

	Radioecology in July 2006, and the submission of 3 abstracts accepted for presentation at scientific conferences. 4) In vitro studies using haematopoietic cells from the bone marrow revealed hypergravity increases TRAP-positive, osteoclast-like cell number whereas irradiation dose-dependently decreases osteoclast-like cell number. 5) Preliminary animal studies revealed rapid structural changes in bone following proton irradiation and altered behavior of cells derived from hindlimb unloaded mice compared to normally loaded controls. . In addition to these scientific accomplishments, three Education and Outreach elements were implemented, consistent with NASA's mission. The first element consisted of recruiting, training, and mentoring students in our radiation research project. Students from various community colleges, colleges, and graduate training programs were recruited and trained during this first year and now are actively conducting research under the mentorship of members of the research project. The student's involvement in this project culminated in the recent presentation by the student of two posters at the Ames Summer Science Day poster session in August, 2006. The second Education and Outreach element engaged a Grade 8-12 web-based project, the Space Settlement Design Contest (http://www.nas.nasa.gov/ which has a 13-year track record of success at Ames Research Center, by asking students to incorporate experiments relevant to radiation in their space settlement designs. The third element was to deliver a lecture on the influence of radiation on tissue degeneration with relevance to space biology and astronaut health for the Radiation Biology Professional Development Course, which was offered to High School teachers at NASA-ARC.
Bibliography Type:	Description: (Last Updated: 06/04/2025)
Abstracts for Journals and Proceedings	Kondo H, Limoli C, Searby ND, Almeida EA, Loftus DJ, Vercoutere W, Hilton D, Globus RK. "Shared oxidative pathway in response to gravity-dependent loading and gamma irradiation of bone marrow-derived skeletal cell progenitors." Radiation processes in living systems: Tissue and organism. 17th Annual NASA Space Radiation Health Investigators Workshop and 4-th International Workshop on Space Radiation Research, Moscow-St Petersburg, June 5-9, 2006. 17th Annual NASA Space Radiation Health Investigators Workshop and 4-th International Workshop on Space Radiation Research, proceedings, June 2006. , Jun-2006